LIBRARY MANAGEMENT SYSTEM

SQL Queries Reference Document

The following queries retrieve book details, member details, issued and returned books, overdue records, and update catalog counts. Run these queries after creating the `Books`, `Members`, and `IssuedBooks` tables and inserting sample data.

1. Display all books

```sql
SELECT * FROM Books;
```

2. Display all members

```sql
SELECT * FROM Members;
```

3. Display all issued-book records

```sql
SELECT * FROM IssuedBooks;
```

4. Display available books

```sql
SELECT BookID, Title, Author, Category, CopiesAvailable FROM Books WHERE CopiesAvailable > 0 ORDER BY Title;
```

5. Display books currently issued to members

```sql
SELECT I.IssueID, B.Title, B.Author, M.MemberName, I.IssueDate, I.DueDate FROM IssuedBooks AS I JOIN Books AS B ON I.BookID = B.BookID JOIN Members AS M ON I.MemberID = M.MemberID WHERE I.ReturnDate IS NULL ORDER BY I.DueDate;
```

6. Display full issue history with book and member details

```sql
SELECT I.IssueID, M.MemberName, B.Title, B.Author, I.IssueDate, I.DueDate, I.ReturnDate FROM IssuedBooks AS I JOIN Books AS B ON I.BookID = B.BookID JOIN Members AS M ON I.MemberID = M.MemberID ORDER BY I.IssueDate;
```

7. Find overdue books

A book is overdue if it has not been returned and the due date has passed.

```
SELECT I.IssueID, M.MemberName, M.Phone, B.Title, I.DueDate, CURRENT_DATE() AS CurrentDate FROM IssuedBooks AS I JOIN Books AS B ON I.BookID = B.BookID JOIN Members AS M ON I.MemberID = M.MemberID WHERE I.ReturnDate IS NULL AND I.DueDate < CURRENT_DATE() ORDER BY I.DueDate;
```

8. Calculate the number of days a book is overdue

```
SELECT I.IssueID, M.MemberName, B.Title, I.DueDate, CURRENT_DATE() AS CurrentDate, DATEDIFF(CURRENT_DATE(), I.DueDate) AS DaysOverdue FROM IssuedBooks AS I JOIN Books AS B ON I.BookID = B.BookID JOIN Members AS M ON I.MemberID = M.MemberID WHERE I.ReturnDate IS NULL AND I.DueDate < CURRENT_DATE() ORDER BY DaysOverdue DESC;
```

9. Calculate fine for overdue books

Fine rate: Rs. 5 per overdue day.

```
SELECT I.IssueID, M.MemberName, B.Title, I.DueDate, DATEDIFF(CURRENT_DATE(), I.DueDate) AS DaysOverdue, DATEDIFF(CURRENT_DATE(), I.DueDate) * 5 AS FineAmount FROM IssuedBooks AS I JOIN Books AS B ON I.BookID = B.BookID JOIN Members AS M ON I.MemberID = M.MemberID WHERE I.ReturnDate IS NULL AND I.DueDate < CURRENT_DATE() ORDER BY FineAmount DESC;
```

10. Count the number of books issued to each member

```
SELECT M.MemberID, M.MemberName, COUNT(I.IssueID) AS BooksIssued FROM Members AS M LEFT JOIN IssuedBooks AS I ON M.MemberID = I.MemberID AND I.ReturnDate IS NULL GROUP BY M.MemberID, M.MemberName ORDER BY BooksIssued DESC;
```

11. Find the most popular books based on number of issues

```
SELECT B.BookID, B.Title, B.Author, COUNT(I.IssueID) AS TimesIssued FROM Books AS B LEFT JOIN IssuedBooks AS I ON B.BookID = I.BookID GROUP BY B.BookID, B.Title, B.Author ORDER BY TimesIssued DESC;
```

12. Find members who have borrowed more than one book currently

```
SELECT M.MemberID, M.MemberName, COUNT(I.IssueID) AS BooksCurrentlyIssued FROM Members AS M JOIN IssuedBooks AS I ON M.MemberID = I.MemberID WHERE I.ReturnDate IS NULL GROUP BY M.MemberID, M.MemberName HAVING COUNT(I.IssueID) > 1 ORDER BY BooksCurrentlyIssued DESC;
```

13. Display books by a particular author

```
SELECT BookID, Title, Category, CopiesAvailable FROM Books WHERE Author = 'R.K. Narayan' ORDER BY Title;
```

14. Display books in a particular category

```
SELECT BookID, Title, Author, CopiesAvailable FROM Books WHERE Category = 'Computer Science' ORDER BY Title;
```

15. Search books by title keyword

```
SELECT BookID, Title, Author, Category, CopiesAvailable FROM Books WHERE Title LIKE '%Database%' ORDER BY Title;
```

16. Display members who have never borrowed a book

```
SELECT M.MemberID, M.MemberName, M.Email FROM Members AS M LEFT JOIN IssuedBooks AS I ON M.MemberID = I.MemberID WHERE I.IssueID IS NULL ORDER BY M.MemberName;
```

17. Display books returned after the due date

```
SELECT I.IssueID, M.MemberName, B.Title, I.DueDate, I.ReturnDate, DATEDIFF(I.ReturnDate, I.DueDate) AS DaysLate FROM IssuedBooks AS I JOIN Books AS B ON I.BookID = B.BookID JOIN Members AS M ON I.MemberID = M.MemberID WHERE I.ReturnDate IS NOT NULL AND I.ReturnDate > I.DueDate ORDER BY DaysLate DESC;
```

18. Count total books, available books, and issued books

```sql
SELECT SUM(CopiesAvailable) AS TotalAvailableCopies, COUNT(*) AS TotalBookTitles FROM Books;
```

19. Display the latest issued books

```sql
SELECT I.IssueID, M.MemberName, B.Title, I.IssueDate, I.DueDate FROM IssuedBooks AS I JOIN Books AS B ON I.BookID = B.BookID JOIN Members AS M ON I.MemberID = M.MemberID ORDER BY I.IssueDate DESC LIMIT 5;
```

20. Update the available copies after a book is issued

Replace 101 with the required BookID.

```sql
UPDATE Books SET CopiesAvailable = CopiesAvailable - 1 WHERE BookID = 101 AND CopiesAvailable > 0;
```

21. Update the available copies after a book is returned

Replace 101 with the required BookID.

```sql
UPDATE Books SET CopiesAvailable = CopiesAvailable + 1 WHERE BookID = 101;
```

22. Mark an issued book as returned

Replace 1 with the required IssueID.

```sql
UPDATE IssuedBooks SET ReturnDate = CURRENT_DATE() WHERE IssueID = 1 AND ReturnDate IS NULL;
```

23. Delete a member who has no issued books

Replace 5 with the required MemberID.

```sql
DELETE FROM Members WHERE MemberID = 5 AND MemberID NOT IN ( SELECT DISTINCT MemberID FROM IssuedBooks );
```